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$f(x) = \frac{4x+5}{6x+7}$

Quotiregel: $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$

Stel $u = 4x + 5$ en $v = 6x + 7$

$u' = \frac{d}{dx}[4x + 5] = 4$

$v' = \frac{d}{dx}[6x + 7] = 6$

$$\frac{df}{dx} = \frac{(6x+7) \cdot \frac{d}{dx}[4x+5] - (4x+5) \cdot \frac{d}{dx}[6x+7]}{(6x+7)^2}$$
$$= \frac{(6x+7) \cdot 4 - (4x+5) \cdot 6}{(6x+7)^2}$$
$$= \frac{4(6x+7) - 6(4x+5)}{(6x+7)^2}$$
$$= \frac{24x + 28 - 24x - 30}{(6x+7)^2}$$
$$= \frac{28 - 30}{(6x+7)^2}$$
$$= \frac{-2}{(6x+7)^2}$$

References